



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

04-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE : ALL STAR BATCH)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	D	B	D	D	A	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	D	C	A	A	C	A	D	D	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	500	2	15	4	3	2	60	5	1	57

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	D	A	D	A	A	B	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	B	B	B	A	D	A	A	C	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1	78	6	3	2	21	106	12	7	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	C	C	A	A	A	D	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	C	D	D	B	C	A	A	D	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	720	96	3125	4	1251	2	4	8	4	96

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)

In capacitor, current lead voltage by $\frac{\pi}{2}$

2. Ans (D)

$$R = \frac{R_1 R_2}{R_1 + R_2} = \frac{\ell}{A} \cdot \frac{\rho_1 \rho_2}{\rho_1 + \rho_2}$$

$$R = \frac{25 \times 10^{-2}}{3 \times 10^{-6}} \times \frac{1.7 \times 2.6 \times 10^{-16}}{4.3 \times 10^{-8}}$$

$$R = 0.858 \text{ m}\Omega$$

3. Ans (D)

$$(A \cos \theta) \hat{B} = A \left(\frac{\vec{A} \cdot \vec{B}}{AB} \right) \hat{B} = \frac{\vec{A} \cdot \vec{B}}{B} \hat{B}$$

$$= \frac{2}{\sqrt{2}} \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}} \right) = \hat{i} + \hat{j}$$

4. Ans (B)

$$W_{\text{Porter}} + W_{\text{mg}} = \Delta K.E. = 0$$

$$W_{\text{Porter}} = -W_{\text{mg}} = -mgh$$

$$= -80 \times 9.8 \times .8 = -627.2 \text{ J}$$

5. Ans (D)

$$T_0 = 2\pi \sqrt{\frac{\ell}{g}}$$

$$\text{New time period } T = 2\pi \sqrt{\frac{\ell/16}{g}} = \frac{2\pi}{4} \sqrt{\frac{\ell}{g}}$$

$$T = \frac{T_0}{4}$$

6. **Ans (D)**

$$r + r' + 90^\circ = 180^\circ \Rightarrow r' = 90 - r = 90 - i$$

$$n_1 \sin i = n_2 \sin r' = n_2 \sin (90 - i)$$

$$n_1 \sin i = n_2 \cos i \Rightarrow \tan i = \frac{n_2}{n_1}$$

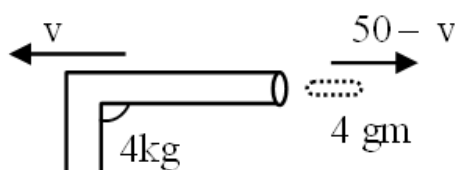
$$\text{Now } \sin C = \frac{n_2}{n_1} = \tan i$$

$$\Rightarrow C = \sin^{-1} (\tan i) = \sin^{-1} (\tan r)$$

7. **Ans (A)**

As temperature increases, domains disintegrate so ferromagnetism decreases and above curie temperature it become paramagnet.

8. **Ans (B)**



By momentum conservation

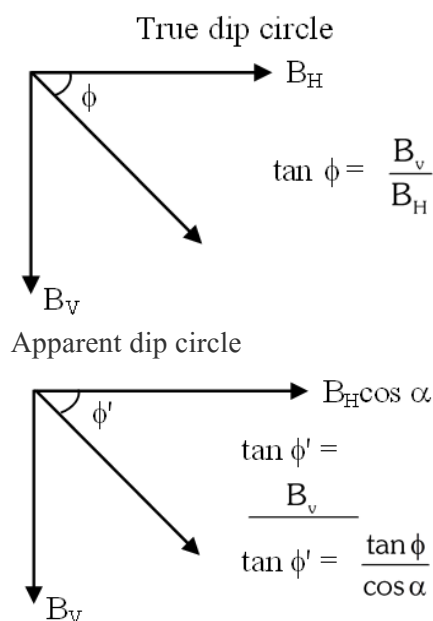
$$4 \times 10^{-3} (50 - v) - 4v = 0$$

$$v = \frac{4 \times 10^{-3} \times 50}{4 + 4 \times 10^{-3}} \approx 0.05 \text{ ms}^{-1}$$

$$\text{Impulse } J = mv = 4 \times .05 = 0.2 \text{ kgms}^{-1}$$

9. **Ans (B)**

If apparent dip circle is at an angle α with true dip circle then



As $\cos \alpha < 1$

Hence true dip (ϕ) is less than apparent dip (ϕ')

10. **Ans (A)**

$$a = \frac{g \sin \theta}{1 + \frac{I}{mR^2}}$$

$$I_{\text{ring}} > I_{\text{solid cylinder}} > I_{\text{solid sphere}}$$

$$\Rightarrow a_{\text{ring}} < a_{\text{solid cylinder}} < a_{\text{solid sphere}}$$

$$\Rightarrow v_{\text{ring}} < v_{\text{solid cylinder}} < v_{\text{solid sphere}}$$

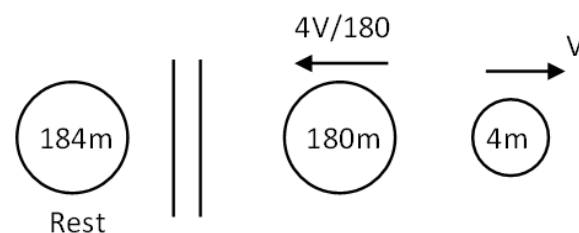
11. **Ans (A)**

$$\text{Band gap} = \frac{hc}{\lambda_0}$$

λ_0 ; threshold wavelength

$$\text{Band gap} = \frac{1242 \text{ eV} \cdot \text{nm}}{621 \text{ nm}} = 2 \text{ eV}$$

12. **Ans (D)**



$$\frac{1}{2} (4m) v^2 + \frac{1}{2} (180m) \left(\frac{4v}{180} \right)^2 = 5.5 \text{ MeV}$$

$$\Rightarrow \frac{1}{2} 4mv^2 \left[1 + 45 \left(\frac{4}{180} \right)^2 \right] = 5.5 \text{ MeV}$$

$$\Rightarrow K.E_\alpha = \frac{5.5}{1 + 45 \cdot \left(\frac{4}{180} \right)^2} \text{ MeV}$$

$$K.E_\alpha = 5.38 \text{ MeV}$$

13. **Ans (C)**

$$KE = e\Delta V$$

$$\lambda_e = \frac{h}{\sqrt{2m_e (e\Delta V)}}$$

$$\lambda_p = \frac{h}{\sqrt{2m_p (e\Delta V)}}$$

$$\Rightarrow \frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$$

14. **Ans (A)**

- (a) For $x_L > x_C$, voltage leads the current
(ii)
(b) For $x_L = x_C$, voltage & current are in same phase
(i)
(c) For $x_L < x_C$, current leads the voltage
(iv)
(d) For resonant frequency $x_L = x_C$, current is maximum
(iii)

15. **Ans (A)**

$$\text{Radius covered } r = \sqrt{2RH_T}$$

$$150 \text{ km} = \sqrt{2 \times (6.5 \times 10^6 \text{ m}) H_T}$$

$$(150 \text{ km} \times 10^3)^2 = 2 \times 6.5 \times 10^6 H_T$$

$$H_T = 1731 \text{ m}$$

$$\text{Population covered} = (\pi r^2)(2000/\text{km}^2)$$

$$= 3.14 \times (150)^2 \times 2000 = 1413 \times 10^5$$

16. **Ans (C)**

As per Equipartition law :

Each degree of freedom contributes

$$\frac{1}{2} k_B T \text{ Average Energy}$$

In monoatomic gas D.O.F. = 3

$$\Rightarrow \text{Average energy} = 3 \times \frac{1}{2} k_B T = \frac{3}{2} k_B T$$

17. **Ans (A)**

$$I_{\text{avg}} = \frac{B_0^2 C}{2\mu_0} \quad \& \quad \frac{1}{\mu_0} = \epsilon_0 C^2$$

$$I = \frac{B_0^2}{2} \epsilon_0 C^3$$

$$B_0 = \sqrt{\frac{2I}{\epsilon_0 C^3}}$$

$$B_0 = 2.77 \times 10^{-8} \text{ T}$$

18. **Ans (D)**

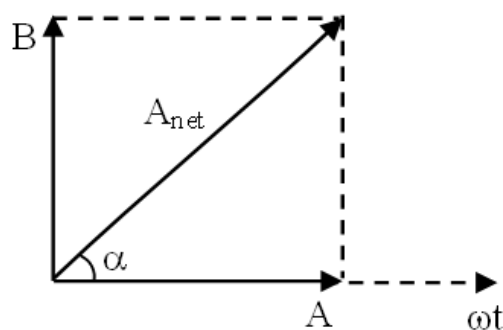
$$x = A \sin \omega t + B \cos \omega t$$

$$v = \frac{dx}{dt} = A\omega \cos \omega t - B\omega \sin \omega t$$

$$\text{At } t = 0, x(0) = B$$

$$v(0) = A\omega$$

$$x = A \sin \omega t + B \sin(\omega t + 90^\circ)$$



$$A_{\text{net}} = \sqrt{A^2 + B^2}$$

$$\tan \alpha = \frac{B}{A} \Rightarrow \cot \alpha = \frac{A}{B}$$

$$\Rightarrow x = \sqrt{A^2 + B^2} \sin(\omega t + \alpha)$$

$$\Rightarrow x = \sqrt{A^2 + B^2} \cos(\omega t - (90 - \alpha))$$

$$x = C \cos(\omega t - \phi)$$

$$\Rightarrow C = \sqrt{A^2 + B^2}$$

$$C = \sqrt{\frac{[v(0)]^2}{\omega^2} + [x(0)]^2}$$

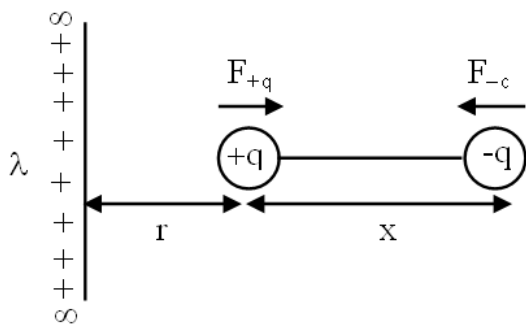
$$\phi = 90 - \alpha$$

$$\tan \alpha = \cot \alpha = \frac{A}{B}$$

$$\Rightarrow \tan \phi = \frac{v(0)}{x(0) \cdot \omega}$$

$$\phi = \tan^{-1} \left(\frac{v(0)}{x(0) \omega} \right)$$

19. Ans (D)



$$r = 10 \text{ mm}, x = 2,$$

$$|\vec{F}_q| = \frac{2k\lambda}{r} \cdot q$$

$$|\vec{F}_{-q}| = \frac{2k\lambda}{r+x} \cdot q$$

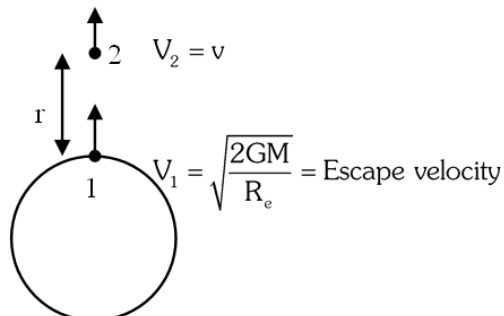
$$\Rightarrow |\vec{F}_{\text{net}}| = \frac{2k\lambda q}{r} - \frac{2k\lambda q}{r+x}$$

$$|\vec{F}_{\text{net}}| = \frac{2k\lambda q \cdot x}{r(r+x)}$$

$$4 = \frac{2 \times 9 \times 10^9 \times 3 \times 10^{-6} \times q \times 2\text{mm}}{10\text{mm} \cdot 12\text{mm}}$$

$$\Rightarrow q = 4.44 \mu\text{C}$$

20. Ans (D)



Applying energy conservation from (1) to (2)

$$\frac{1}{2} m \cdot \left(\frac{2GM}{R_e} \right) - \frac{GMm}{R_e} = \frac{1}{2} mv^2 - \frac{GMm}{R+r}$$

$$\Rightarrow \frac{1}{2} mv^2 = \frac{GMm}{R+r}$$

$$\Rightarrow v = \sqrt{\frac{2GM}{R+r}} = \frac{dr}{dt}$$

$$\Rightarrow \sqrt{2GM} \int_0^t dt = \int_{R_e}^{R_e+h} (\sqrt{R+r}) dr$$

$$\sqrt{2GM} \cdot t = \frac{2}{3} \left[(R+r)^{3/2} \right]_{R_e}^{R_e+h}$$

$$t = \frac{2}{3} \sqrt{\frac{R_e^3}{2GM}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$$

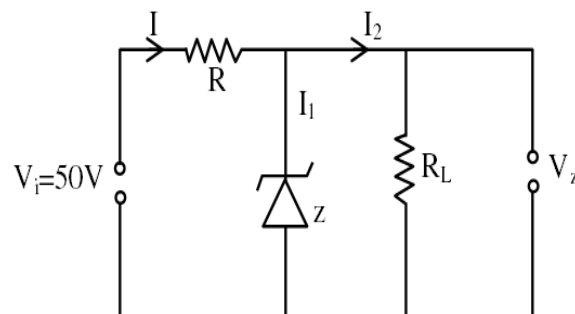
$$\frac{GM}{R_e^2} = g$$

$$t = \frac{1}{3} \sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$$

PART-1 : PHYSICS

SECTION-II

1. Ans (500)



Voltage across $R_L = 5\text{V}$

$$\Rightarrow i_2 = \frac{5}{R_L}$$

Also voltage across $R = 50 - 5 = 45 \text{ volt}$

$$\text{By } v = iR \Rightarrow R = \frac{v}{i} = \frac{45}{i_1 + i_2}$$

$$R = \frac{45}{90\text{mA} + \frac{5}{R_L}}$$

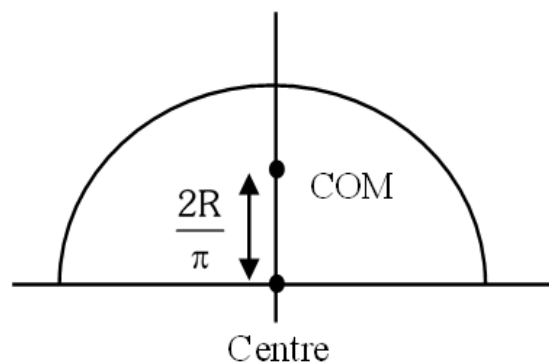
Current in zener diode is maximum when

$$R_L \rightarrow \infty (i_2 \rightarrow 0 \text{ and } i_1 = i)$$

$$\text{So } R = \frac{45}{90\text{mA}} = 500\Omega$$

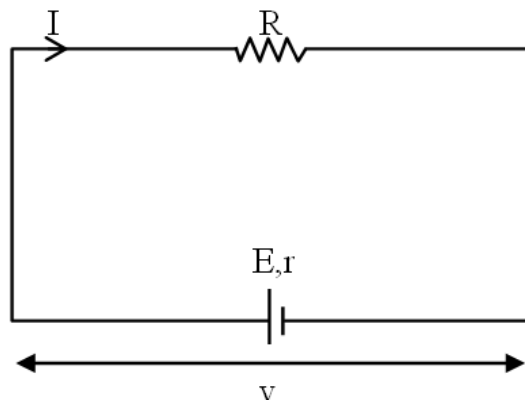
2. Ans (2)

COM of semi-circular ring is at $\frac{2R}{\pi}$



Distance from centre $\Rightarrow x = 2$

3. Ans (15)



$$\text{Terminal voltage } V = IR = \frac{ER}{R+r}$$

$$1^{\text{st}} \rightarrow 1.25 = \frac{E(5)}{5+r} \quad \dots(i)$$

$$2^{\text{nd}} \rightarrow 1 = \frac{E(2)}{2+r} \quad \dots(ii)$$

By (i) and (ii)

$$r = 1\Omega, E = \frac{3}{2}V = \frac{15}{10}\text{ volt}$$

$$\Rightarrow x = 15$$

4. Ans (4)

Electric flux density

$$(\vec{D}) = \frac{\text{charge}}{\text{Area}} \times \hat{r} = \frac{Q}{4\pi r^2} \hat{r} = \epsilon_0 \left(\frac{Q}{4\pi \epsilon_0 r^2} \hat{r} \right)$$

$$\Rightarrow \vec{E} = \frac{\vec{D}}{\epsilon_0} = \frac{e^{-x} \sin y \hat{i} - e^{-x} \cos y \hat{j} + 2z \hat{k}}{\epsilon_0}$$

Also by Gauss's law

$$\frac{\rho}{\epsilon_0} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \vec{E}$$

$$= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot \frac{\vec{D}}{\epsilon_0}$$

$$\Rightarrow \rho = \frac{\partial}{\partial x} (e^{-x} \sin y) + \frac{\partial}{\partial y} (-e^{-x} \cos y) + \frac{\partial}{\partial z} (2z)$$

$$\rho = -e^{-x} \sin y + e^{-x} \sin y + 2$$

$$\text{At origin } \rho = -e^{-0} \sin 0 + e^{-0} \sin 0 + 2$$

$$\rho = 2C/m^3$$

$$\text{Charge} = \rho \times \text{volume} = 2 \times 2 \times 10^{-9} = 4 \times 10^{-9}$$

$$= 4nC$$

5. Ans (3)

$$\text{Direction of P } \hat{v}_1 = \pm \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} = \pm \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$$

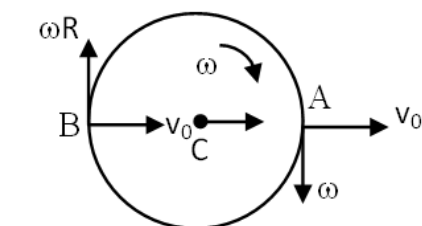
$$\text{Direction of Q } \hat{v}_2 = \pm \frac{\vec{A} \times \vec{C}}{|\vec{A} \times \vec{C}|} = \pm \frac{2\hat{k}}{2} = \pm \hat{k}$$

Angle between $\hat{v}_1$ and $\hat{v}_2$

$$\frac{\hat{v}_1 \cdot \hat{v}_2}{|\hat{v}_1| |\hat{v}_2|} = \frac{\pm 1/\sqrt{3}}{(1)(1)} = \pm \frac{1}{\sqrt{3}}$$

$$\Rightarrow x = 3$$

6. Ans (2)



For no slipping $v_0 = \omega R$

$$\text{Now } v_A = v_B = \sqrt{v_0^2 + (\omega R)^2}$$

$$= \sqrt{2}v_0$$

$$\Rightarrow x = 2$$

7. Ans (60)

At minimum deviation $r_1 = r_2 = \frac{A}{2}$

Also given $i = 2r_1 = A$

Now $1. \sin i = \sqrt{3} \sin r_1$

$$1 \sin A = \sqrt{3} \sin \frac{A}{2}$$

$$\Rightarrow 2 \sin \frac{A}{2} \cos \frac{A}{2} = \sqrt{3} \sin \frac{A}{2}$$

$$\Rightarrow \cos \frac{A}{2} = \frac{\sqrt{3}}{2} \Rightarrow \frac{A}{2} = 30^\circ$$

$$\Rightarrow A = 60^\circ$$

8. Ans (5)

$$\text{Elastic energy} = \frac{Y}{2} (\text{strain})^2 \times \text{Area} \times \text{length}$$

$\Rightarrow$ Elastic energy per unit length

$$= \frac{Y}{2} (\text{strain})^2 \times \text{Area}$$

$$\left(\text{strain} = \frac{\Delta \ell}{\ell} = \alpha \Delta T = 10^{-5} \times 10 = 10^{-4} \right)$$

$$= \frac{10^{11}}{2} \times (10^{-4})^2 \times 10^{-2} = 5J/m$$

9. Ans (1)

$$T = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow g = \frac{4\pi^2 \ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

$$\Delta T = \frac{\text{least count of time } (\Delta T_0)}{\text{number of oscillations}(n)}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T_0}{nT}$$

As $\Delta \ell$ and ΔT_0 are same for all observations so

$\frac{\Delta g}{g}$ is minimum for highest value of ℓ , n and T

$\Rightarrow$ Minimum percentage error in g is for student number-1

10. Ans (57)

By newton's law of cooling (with approximation)

$$\frac{\Delta T}{\Delta t} = -C (T_{\text{avg}} - T_s)$$

$$1^{\text{st}} \frac{-10^\circ\text{C}}{5 \text{ min}} = -C (70^\circ\text{C} - 25^\circ\text{C})$$

$$\Rightarrow C = \frac{2}{45} \text{ min}^{-1}$$

$$2^{\text{nd}} \frac{T - 65}{5 \text{ min}} = -C \left(\frac{T + 65}{2} - 25 \right)$$

$$= - \left(\frac{2}{45} \right) \left(\frac{T + 15}{2} \right)$$

$$\Rightarrow 9(T - 65) = -(T + 15)$$

$$\Rightarrow 10T = 570$$

$$\Rightarrow T = 57^\circ\text{C}$$

Alternate Solution :

Newton's law of cooling (without approximation)

$$T_P - T_S = (T_i - T_S) e^{-Ct}$$

$$1^{\text{st}} \quad 65 - 25 = (75 - 25)e^{-5C} \Rightarrow e^{-5C} = \frac{4}{5}$$

$$2^{\text{nd}} \quad T - 25 = (65 - 25)e^{-5C} = 40 \times \frac{4}{5} = 32$$

$$T = 57^\circ\text{C}$$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (D)

On heating concentration of O_2 in water decreases. So boiling water and water at 80°C having less O_2 concentration. Polluted water also having less O_2 concentration. So water at 4°C having maximum O_2 concentration.

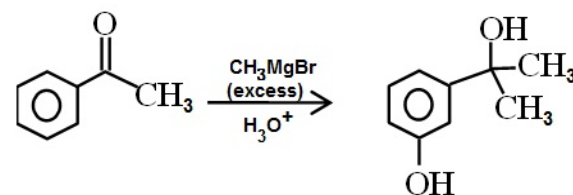
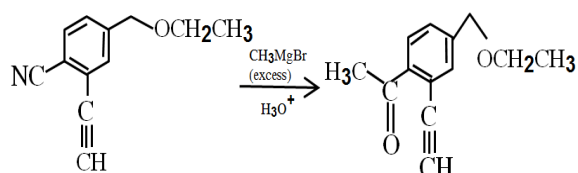
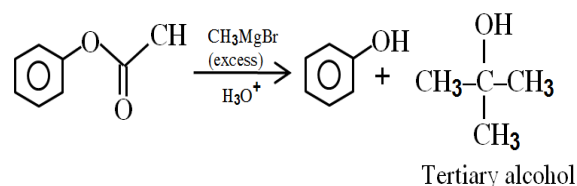
2. Ans (B)

At the time, he proposed the periodic table but structure of atom was unknown.

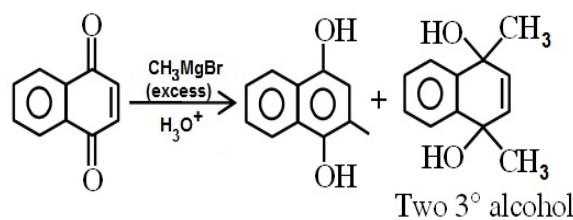
3. Ans (D)

Reduced pressure distillation or vacuum distillation is used for the purification of high boiling organic liquids which decomposes at or below their boiling point.

4. Ans (A)

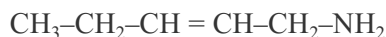


Phenolic $-\text{OH}$ group and tertiary alcohol is present thus two functional groups are present in the product



Since the given question is single correct choice the best appropriate option is (A)

5. **Ans (D)**



No conjugation thus resonance is not possible.

6. **Ans (A)**

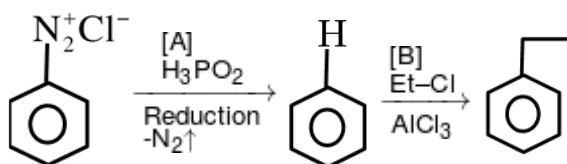
(a) 'Ba' having outer electronic configuration $6s^2$.

(b) CaC_2O_4 is water insoluble

(c) 'Li' is soluble in organic solvents

(d) NaOH is strong Monoacidic base among given.

7. **Ans (A)**



8. **Ans (B)**

^1_1H and ^2_1H are stable while ^3_1H is radioactive.

9. **Ans (C)**

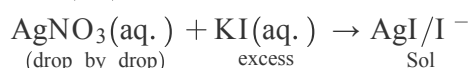
(a) SF_4 – sp^3d hybridisation

(b) IF_5 – sp^3d^2 hybridisation

(c) NO_2^+ – sp hybridisation

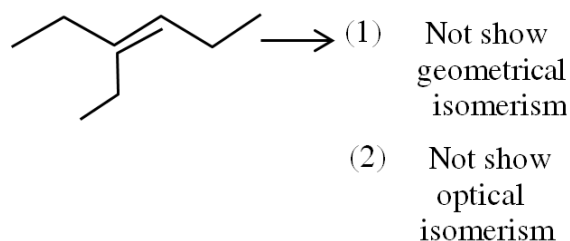
(d) NH_4^+ – sp^3 hybridisation

10. **Ans (A)**



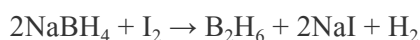
11. **Ans (C)**

3-Ethylhex-3-ene will not show stereo isomerism its diagram is.



12. **Ans (B)**

Diborane is prepared by the reaction of NaBH_4 with I_2 .



In diborane, 'B' is sp^3 hybrid, it is Non-planar and two $3\text{c}-2\text{e}^-$ bonds are present.

13. **Ans (B)**

Among 15^{th} group hydrides, BiH_3 is strongest reducing agent.

14. **Ans (B)**

(a)	Chloroprene	(i)	
(b)	Neoprene	(ii)	
(c)	Acrylonitrile	(iii)	$\text{CH}_2=\text{CH}-\text{CN}$
(d)	Isoprene	(iv)	

15. **Ans (A)**

$\text{Cu}^{2+} : [\text{Ar}]3\text{d}^9 4\text{s}^0$
 $\text{Cr}^{3+} : [\text{Ar}]3\text{d}^3 4\text{s}^0$
 $\text{Sc}^+ : [\text{Ar}]3\text{d}^1 4\text{s}^1$

All are coloured and paramagnetic due to presence of unpaired electrons

16. **Ans (D)**

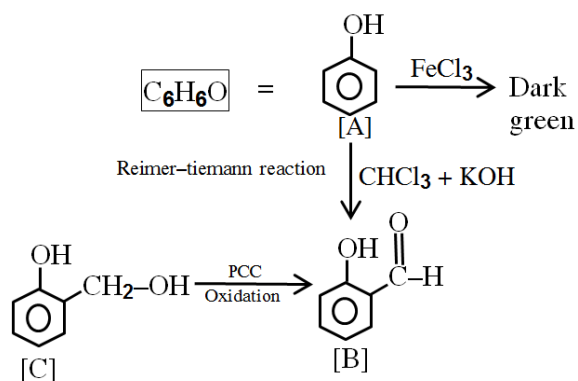
Vitamine- B_1 is also known as Thiamine while vitamin B-6 is known as Pyridoxine

17. **Ans (A)**

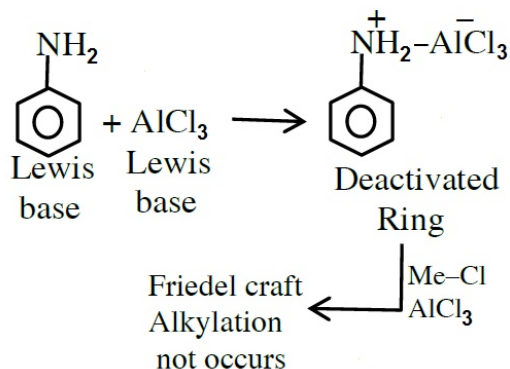
Pb and Ag commonly exist in the form of sulphide ore like PbS (galena) and Ag_2S (Argentite)

'Al' is mainly found in the form of oxide ore whereas 'Mg' is found in the form of halide ore.

18. **Ans (A)**



19. Ans (C)



- (1) Aniline is lewis base give acid base reaction with AlCl_3 and form Anilinium ion
 (2) Anilinium ion has strongest deactivated ring so further Friedel craft Alkylation not occurs.

20. Ans (A)

$$T_f - T'_f = i K_f \cdot m$$

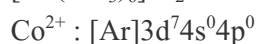
For minimum T'_f
 'i' should be maximum.



PART-2 : CHEMISTRY

SECTION-II

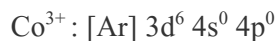
1. Ans (1)



For this complex $\Delta_0 < \text{P.E.}$, so pairing of electron does not take place.

$sp^3 d^2$ hybridisation

Total 3 unpaired electrons are present.

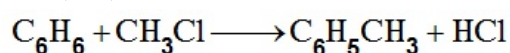


$d^2 sp^3$ hybridisation

NH_3 acts as SFL because $\Delta_0 > \text{P.E.}$

So here all electrons becomes paired.

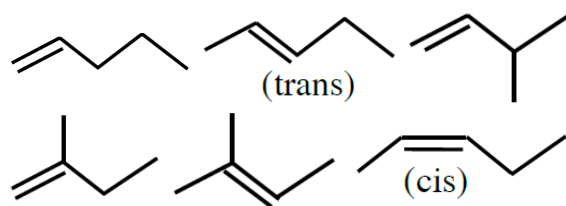
2. Ans (78)



$$\frac{10}{78} \left(\frac{10}{78} \times 92 \right) \text{ gm} \Rightarrow$$

$$\frac{A_y}{T_y} = \% \text{ yield} = \frac{9.2}{920} \times 78 \times 100 \Rightarrow 78\%$$

3. Ans (6)



4. Ans (3)

$$E = E^\circ - \frac{0.059}{2} \log \frac{[\text{Cu}^{+2}]}{[\text{Ag}^+]^2}$$

$$= 2.97 - \frac{0.059}{2} \log \frac{0.25}{(10^{-3})^2} = 2.81 \text{ V}$$

5. Ans (2)

$$K_C = \frac{K_P}{RT} = \frac{47.9}{0.083 \times 288} = 2$$

6. Ans (21)

1 mol graphite = 12 gm C

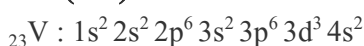
$$\text{Ans.} = \frac{248}{12} = 20.67 \text{ kJ/gm heat evolved}$$

7. Ans (106)

$$d \left(\frac{\text{gm}}{\text{cc}} \right) = \frac{4 \times \frac{M}{N_A}}{(a \text{ cm})^3}$$

$$7.62 = \frac{4 \times M / 6.022 \times 10^{23}}{(0.4518 \times 10^{-7} \text{ cm})^3} \Rightarrow M = 105.8 \text{ g/mol}$$

8. Ans (12)



Number of electrons in p-orbitals is equal to 12.00

9. Ans (7)

$$K = \frac{2.303}{t} \log \frac{[\text{N}_2\text{O}_5]_0}{[\text{N}_2\text{O}_5]_t}$$

$$= \frac{2.303}{60} \log \frac{2.4}{1.6}$$

$$= 6.76 \times 10^{-3} \text{ min}^{-1} \approx 7 \times 10^{-3} \text{ min}^{-1}$$

10. Ans (4)

$$[\text{Glucose}] = \frac{C(\text{gm/l})}{M(\text{gm/mol})} = \frac{0.72}{180} = 4 \times 10^{-3} \text{ M}$$

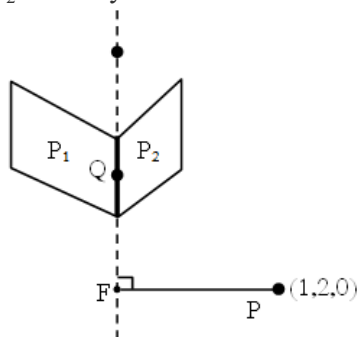
PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

$$P_1 : x - y + 2z = 2$$

$$P_2 : 2x + y - 3 = 2$$



Let line of Intersection of planes P_1 and P_2 cuts xy plane in point Q .

$\Rightarrow$ z -coordinate of point Q is zero

$$\Rightarrow \begin{cases} x - y = 2 \\ \text{and } 2x + y = 2 \end{cases} \Rightarrow x = \frac{4}{3}, y = \frac{-2}{3}$$

$$\Rightarrow Q \left(\frac{4}{3}, \frac{-2}{3}, 0 \right)$$

Vector parallel to the line of intersection

$$\vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 2 & 1 & -1 \end{vmatrix} = -\hat{i} + 5\hat{j} + 3\hat{k}$$

Equation of Line of intersection

$$\frac{x - \frac{4}{3}}{-1} = \frac{y + \frac{2}{3}}{5} = \frac{z - 0}{3} = \lambda \text{ (say)}$$

Let coordinates of foot of perpendicular be

$$F \left(-\lambda + \frac{4}{3}, 5\lambda - \frac{2}{3}, 3\lambda \right)$$

$$\vec{PF} = \left(-\lambda + \frac{1}{3} \right) \hat{i} + \left(5\lambda - \frac{8}{3} \right) \hat{j} + (3\lambda) \hat{k}$$

$$\vec{PF} \cdot \vec{a} = 0$$

$$\Rightarrow \lambda - \frac{1}{3} + 25\lambda - \frac{40}{3} + 9\lambda = 0$$

$$\Rightarrow 35\lambda = \frac{41}{3} \Rightarrow \lambda = \frac{41}{105}$$

$$\text{Now, } \alpha = -\lambda + \frac{4}{3}, \beta = 5\lambda - \frac{2}{3}, \gamma = 3\lambda$$

$$\Rightarrow \alpha + \beta + \gamma = 7\lambda + \frac{2}{3}$$

$$= 7 \left(\frac{41}{105} \right) + \frac{2}{3}$$

$$= \frac{51}{15}$$

$$\Rightarrow 35(\alpha + \beta + \gamma) = \frac{51}{15} \times 35 = 119$$

2. **Ans (A)**

$$S_{10} = 530 \Rightarrow \frac{10}{2} \{2a + 9d\} = 530$$

$$\Rightarrow 2a + 9d = 106 \dots (1)$$

$$\text{and } S_5 = 140 \Rightarrow \frac{5}{2} \{2a + 4d\} = 140$$

$$\Rightarrow 2a + 4d = 56 \dots (2)$$

$$\Rightarrow 5d = 50 \Rightarrow d = 10 \Rightarrow a = 8$$

$$\text{Now, } S_{20} - S_6 = \frac{20}{2} \{2a + 19d\} - \frac{6}{2} \{2a + 5d\}$$

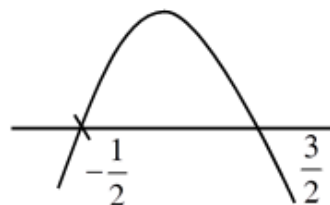
$$= 14a + 175d$$

$$= (14 \times 8) + (175 \times 10)$$

$$= 1862$$

3. **Ans (C)**

$$f'(x) = \begin{cases} -4x^2 + 4x + 3 & x > 0 \\ 3e^x (1 + x) & x \leq 0 \end{cases}$$



$$\text{For } x > 0, f'(x) = -4x^2 + 4x + 3$$

$$f(x) \text{ is increasing in } \left(-\frac{1}{2}, \frac{3}{2} \right)$$

$$\text{For } x \leq 0, f'(x) = 3e^x (1 + x)$$

$$f'(x) > 0 \forall x \in (-1, 0)$$

$$\Rightarrow f(x) \text{ is increasing in } (-1, 0)$$

So, in complete domain, $f(x)$ is increasing in

$$\left(-1, \frac{3}{2} \right)$$

4. **Ans (C)**

$$\frac{dy}{dx} + 2\sin^2 x = 1 + y \cos 2x$$

$$\Rightarrow \frac{dy}{dx} + (-\cos 2x)y = \cos 2x$$

$$I.F. = e^{\int -\cos 2x dx} = e^{-\frac{\sin 2x}{2}}$$

Solution of D.E.

$$y \left(e^{-\frac{\sin 2x}{2}} \right) = \int (\cos 2x) \left(e^{-\frac{\sin 2x}{2}} \right) dx + c$$

$$\Rightarrow y \left(e^{-\frac{\sin 2x}{2}} \right) = -e^{-\frac{\sin 2x}{2}} + c$$

Given

$$y \left(\frac{\pi}{4} \right) = 0$$

$$\Rightarrow 0 = -e^{-\frac{1}{2}} + c \Rightarrow \boxed{c = e^{-\frac{1}{2}}}$$

$$\Rightarrow y \left(e^{-\frac{\sin 2x}{2}} \right) = -e^{-\frac{\sin 2x}{2}} + e^{-\frac{1}{2}}$$

at $x = 0$

$$y = -1 + e^{-\frac{1}{2}}$$

$$\Rightarrow y(0) = -1 + e^{-\frac{1}{2}} \Rightarrow (y(0) + 1)^2 = e^{-1}$$

5. **Ans (A)**

$$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad |A| = ad - bc$$

Total case = 6^4

For non-singular matrix $|A| \neq 0 \Rightarrow ad - bc \neq 0$

$\Rightarrow ad \neq bc$

And a, b, c, d are all different numbers in the set $\{1, 2, 3, 4, 5, 6\}$

Now for $ad = bc$

(i) $6 \times 1 = 2 \times 3$

$$\Rightarrow \left. \begin{array}{l} a = 6, b = 2, c = 3, d = 1 \\ \text{or } a = 1, b = 2, c = 3, d = 6 \\ \vdots \\ \vdots \end{array} \right\} 8 \text{ such cases}$$

(ii) $6 \times 2 = 3 \times 4$

$$\Rightarrow \left. \begin{array}{l} a = 6, b = 3, c = 4, d = 2 \\ \text{or } a = 2, b = 3, c = 4, d = 6 \\ \vdots \\ \vdots \end{array} \right\} 8 \text{ such cases}$$

favourable cases

$$= {}^6C_4 [4 - 16]$$

required probability

$$= \frac{{}^6C_4 [4 - 16]}{6^4} = \frac{43}{162}$$

6. **Ans (A)**

$$\vec{a} = \lambda \vec{b} + \mu \vec{c} = \hat{i} (2\lambda + \mu) + \hat{j} (\lambda - \mu) + \hat{k} (\lambda + \mu)$$

$$\vec{a} \cdot \vec{d} = 0 = 3(2\lambda + \mu) + 2(\lambda - \mu) + 6(\lambda + \mu)$$

$$\Rightarrow 14\lambda + 7\mu = 0 \Rightarrow \mu = -2\lambda$$

$$\Rightarrow \vec{a} = (0) \hat{i} - 3\lambda \hat{j} + (-\lambda) \hat{k}$$

$$\Rightarrow |\vec{a}| = \sqrt{10} \quad |\lambda| = \sqrt{10} \Rightarrow |\lambda| = 1$$

$$\Rightarrow \lambda = 1 \text{ or } -1$$

$$\left[\begin{array}{ccc} \vec{a} & \vec{b} & \vec{c} \end{array} \right] = 0$$

$$\left[\begin{array}{ccc} \vec{a} & \vec{b} & \vec{c} \end{array} \right] + \left[\begin{array}{ccc} \vec{a} & \vec{b} & \vec{d} \end{array} \right] + \left[\begin{array}{ccc} \vec{a} & \vec{c} & \vec{d} \end{array} \right] = \left[\begin{array}{ccc} \vec{a} & \vec{b} + \vec{c} & \vec{d} \end{array} \right]$$

$$= \begin{vmatrix} 0 & -3\lambda & \lambda \\ 3 & 0 & 2 \\ 3 & 2 & 6 \end{vmatrix}$$

$$= 3\lambda(12) + \lambda(6) = 42\lambda = -42$$

7. **Ans (A)**

$$I = \int_0^{100\pi} \frac{\sin^2 x}{e^{\frac{x}{\pi}}} dx = 100 \int_0^{\pi} \frac{\sin^2 x}{e^{\frac{x}{\pi}}} dx$$

$$100 \int_0^{\pi} e^{-\frac{x}{\pi}} \frac{(1 - \cos 2x)}{2} dx$$

$$= 50 \left\{ \int_0^{\pi} e^{-\frac{x}{\pi}} dx - \int_0^{\pi} e^{-\frac{x}{\pi}} \cos 2x dx \right\}$$

$$I_1 = \int_0^{\pi} e^{-\frac{x}{\pi}} dx = \left[-\pi e^{-\frac{x}{\pi}} \right]_0^{\pi} = \pi (1 - e^{-1})$$

$$I_2 = \int_0^{\pi} e^{-\frac{x}{\pi}} \cos 2x dx$$

$$= -\pi e^{-\frac{x}{\pi}} \cos 2x \Big|_0^{\pi} - \int_0^{\pi} -\pi e^{-\frac{x}{\pi}} (-2 \sin 2x) dx$$

$$= \pi (1 - e^{-1}) - 2\pi \int_0^{\pi} e^{-\frac{x}{\pi}} \sin 2x dx$$

$$= \pi (1 - e^{-1}) - 2\pi \left\{ -\pi e^{-\frac{x}{\pi}} \sin 2x \Big|_0^{\pi} - \int_0^{\pi} -\pi e^{-\frac{x}{\pi}} 2 \cos 2x dx \right\}$$

$$= \pi (1 - e^{-1}) - 4\pi^2 I_2$$

$$\Rightarrow I_2 = \frac{\pi (1 - e^{-1})}{1 + 4\pi^2}$$

$$\therefore I = 50 \left\{ \pi (1 - e^{-1}) - \frac{\pi (1 - e^{-1})}{1 + 4\pi^2} \right\}$$

$$= \frac{200 (1 - e^{-1}) \pi^3}{1 + 4\pi^2}$$

8. Ans (D)

$$(1) \vec{a} \times \left((\vec{b} + \vec{c}) \times (\vec{b} - \vec{c}) \right) \\ = \vec{a} \times (-\vec{b} \times \vec{c} + \vec{c} \times \vec{b}) = -2(\vec{a} \times (\vec{b} \times \vec{c})) \\ = -2(\vec{a} \times \vec{a}) = \vec{0}$$

(2) Projection of $\vec{a}$ on $\vec{b} \times \vec{c}$

$$= \frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{|\vec{b} \times \vec{c}|} = \frac{\vec{a} \cdot \vec{a}}{|\vec{a}|} = |\vec{a}| = 2$$

$$(3) [\vec{a} \vec{b} \vec{c}] + [\vec{c} \vec{a} \vec{b}] = 2[\vec{a} \vec{b} \vec{c}] = 2\vec{a} \cdot (\vec{b} \times \vec{c}) \\ = 2\vec{a} \cdot \vec{a} = 2|\vec{a}|^2 = 8$$

$$(4) \vec{a} \times \vec{b} = \vec{c} \text{ and } \vec{b} \times \vec{c} = \vec{a} \\ \Rightarrow \vec{a}, \vec{b}, \vec{c} \text{ are mutually } \perp \text{ vectors.}$$

$$\therefore |\vec{a} \times \vec{b}| = |\vec{c}| \Rightarrow |\vec{a}||\vec{b}| = |\vec{c}| \Rightarrow |\vec{b}| = \frac{|\vec{c}|}{2}$$

$$\text{Also, } |\vec{b} \times \vec{c}| = |\vec{a}| \Rightarrow |\vec{b}||\vec{c}| = 2$$

$$\Rightarrow |\vec{c}| = 2 \text{ \& } |\vec{b}| = 1$$

$$\left| 3\vec{a} + \vec{b} - 2\vec{c} \right|^2 \\ = (3\vec{a} + \vec{b} - 2\vec{c}) \cdot (3\vec{a} + \vec{b} - 2\vec{c}) \\ = 9|\vec{a}|^2 + |\vec{b}|^2 + 4|\vec{c}|^2 \\ = (9 \times 4) + 1 + (4 \times 4) \\ = 36 + 1 + 16 = 53$$

9. Ans (D)

$$x + y + z = 6 \quad \dots(i)$$

$$3x + 5y + 5z = 26 \quad \dots(ii)$$

$$x + 2y + \lambda z = \mu \quad \dots(iii)$$

$$5 \times (i) - (ii) \Rightarrow 2x = 4 \Rightarrow x = 2$$

$\therefore$ from (i) and (iii)

$$y + z = 4 \quad \dots(iv)$$

$$2y + \lambda z = \mu - 2 \quad \dots(v)$$

$$(v) - 2 \times (iv)$$

$$\Rightarrow (\lambda - 2)z = \mu - 10$$

$$\Rightarrow z = \frac{\mu - 10}{\lambda - 2} \quad \&$$

$$y = 4 - \frac{\mu - 10}{\lambda - 2}$$

$\therefore$ For no solution $\lambda = 2$ and $\mu \neq 10$.

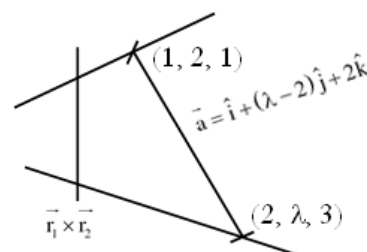
10. Ans (A)

$$L_1: \frac{(x-1)}{2} = \frac{(y-2)}{1} = \frac{(z-1)}{3}$$

$$\vec{r}_1 = 2\hat{i} + \hat{j} + 3\hat{k}$$

$$L_2: \frac{(x-2)}{1} = \frac{(y-\lambda)}{2} = \frac{(z-3)}{4}$$

$$\vec{r}_2 = \hat{i} + 2\hat{j} + 4\hat{k}$$



Shortest distance = Projection of $\vec{a}$ on $\vec{r}_1 \times \vec{r}_2$

$$= \frac{|\vec{a} \cdot (\vec{r}_1 \times \vec{r}_2)|}{|\vec{r}_1 \times \vec{r}_2|}$$

$$|\vec{a} \cdot (\vec{r}_1 \times \vec{r}_2)| = \begin{vmatrix} 1 & \lambda-2 & 2 \\ 2 & 1 & 3 \\ 1 & 2 & 4 \end{vmatrix} = |14 - 5\lambda|$$

$$|\vec{r}_1 \times \vec{r}_2| = \sqrt{38}$$

$$\therefore \frac{1}{\sqrt{38}} = \frac{|14 - 5\lambda|}{\sqrt{38}}$$

$$\Rightarrow |14 - 5\lambda| = 1$$

$$\Rightarrow 14 - 5\lambda = 1 \text{ or } 14 - 5\lambda = -1$$

$$\Rightarrow \lambda = \frac{13}{5} \text{ or } 3$$

$\therefore$ Integral value of $\lambda = 3$.

11. **Ans (D)**

$$(1) (p \rightarrow q) \vee (\sim q \rightarrow p)$$

$$= (\sim p \vee q) \vee (q \vee p)$$

$$= (\sim p \vee p) \vee q$$

$$= t \vee q = t$$

$$(2) (q \rightarrow p) \vee (\sim q \rightarrow p)$$

$$= (\sim q \vee p) \vee (q \vee p)$$

$$= (\sim q \vee q) \vee p$$

$$= t \vee p = t$$

$$(3) (p \rightarrow \sim q) \vee (\sim q \rightarrow p)$$

$$= (\sim p \vee \sim q) \vee (q \vee p)$$

$$= (\sim p \vee p) \vee (\sim q \vee q)$$

$$= t \vee t = t$$

$$(4) (\sim q \rightarrow q) \vee (\sim q \rightarrow p)$$

$$= (p \vee q) \vee (q \vee p)$$

$$= (p \vee p) \vee (q \vee p)$$

$$= p \vee q$$

Which is not a tautology

12. **Ans (C)**

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{Let } X = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$AX = \begin{bmatrix} a_{11} + a_{12} + a_{13} \\ a_{21} + a_{22} + a_{23} \\ a_{31} + a_{32} + a_{33} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow AX = X$$

Replace X by AX

$$A^2X = AX = X$$

Replace X by AX

$$A^3X = AX = X$$

$$\text{Let } A^3 = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

$$A^3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 + x_3 \\ y_1 + y_2 + y_3 \\ z_1 + z_2 + z_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Sum of all the element = 3

13. **Ans (D)**

$$[e^x]^2 + [e^x + 1] - 3 = 0$$

$$\Rightarrow [e^x]^2 + [e^x] + 1 - 3 = 0$$

$$\text{Let } [e^x] = t$$

$$\Rightarrow t^2 + t - 2 = 0$$

$$\Rightarrow t = -2, 1$$

$$[e^x] = -2 \text{ (Not possible)}$$

$$\text{or } [e^x] = 1 \therefore 1 \leq e^x < 2$$

$$\Rightarrow \ln(1) \leq x < \ln(2)$$

$$\Rightarrow 0 \leq x < \ln(2)$$

$$\Rightarrow x \in [0, \ln 2)$$

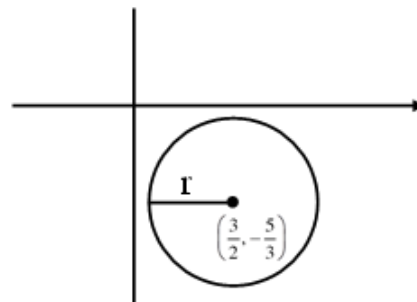
14. **Ans (D)**

$$S : 36x^2 + 36y^2 - 108x + 120y + C = 0$$

$$\Rightarrow x^2 + y^2 - 3x + \frac{10}{3}y + \frac{C}{36} = 0$$

$$\text{Centre} \equiv (-g, -f) \equiv \left(\frac{3}{2}, -\frac{10}{6}\right)$$

$$\text{radius} = r = \sqrt{\frac{9}{4} + \frac{100}{36} - \frac{C}{36}}$$



Now,

$$\Rightarrow r < \frac{3}{2}$$

$$\Rightarrow \frac{9}{4} + \frac{100}{36} - \frac{C}{36} < \frac{9}{4}$$

$$\Rightarrow C > 100 \quad \dots(1)$$

Now point of intersection of $x - 2y = 4$ and

$2x - y = 5$ is $(2, -1)$, which lies inside the circle S

$$\therefore S(2, -1) < 0$$

$$\Rightarrow (2)^2 + (-1)^2 - 3(2) + \frac{10}{3}(-1) + \frac{C}{36} < 0$$

$$\Rightarrow 4 + 1 - 6 - \frac{10}{3} + \frac{C}{36} < 0$$

$$\boxed{C < 156} \quad \dots(2)$$

From (1) & (2)

$$\boxed{100 < C < 156} \text{ Ans.}$$

15. Ans (B)

$$z^2 + 3\bar{z} = 0$$

$$\text{Put } z = x + iy$$

$$\Rightarrow x^2 - y^2 + 2ixy + 3(x - iy) = 0$$

$$\Rightarrow (x^2 - y^2 + 3x) + i(2xy - 3y) = 0 + i0$$

$$\therefore x^2 - y^2 + 3x = 0 \quad \dots(1)$$

$$2xy - 3y = 0 \quad \dots(2)$$

$$x = \frac{3}{2}, y = 0$$

$$\text{Put } x = \frac{3}{2} \text{ in equation (1)}$$

$$\frac{9}{4} - y^2 + \frac{9}{2} = 0$$

$$y^2 = \frac{27}{4} \Rightarrow y = \pm \frac{3\sqrt{3}}{2}$$

$$\therefore (x, y) = \left(\frac{3}{2}, \frac{3\sqrt{3}}{2}\right), \left(\frac{3}{2}, -\frac{3\sqrt{3}}{2}\right)$$

$$\text{Put } y = 0 \Rightarrow x^2 - 0 + 3x = 0$$

$$x = 0, -3$$

$$\therefore (x, y) = (0, 0), (-3, 0)$$

$$\therefore \text{No of solutions} = n = 4$$

$$\sum_{k=0}^{\infty} \left(\frac{1}{n^k}\right) = \sum_{k=0}^{\infty} \left(\frac{1}{4^k}\right)$$

$$= \frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots$$

$$= \frac{1}{1 - \frac{1}{4}} = \frac{4}{3}$$

16. Ans (C)

$$\sin^7 x \leq \sin^2 x \leq 1 \quad \dots(1)$$

$$\text{and } \cos^7 x \leq \cos^2 x \leq 1 \quad \dots(2)$$

$$\text{also } \sin^2 x + \cos^2 x = 1$$

$$\Rightarrow \text{equality must hold for (1) \& (2)}$$

$$\Rightarrow \sin^7 x = \sin^2 x \& \cos^7 x = \cos^2 x$$

$$\Rightarrow \sin x = 0 \& \cos x = 1$$

or

$$\cos x = 0 \& \sin x = 1$$

$$\Rightarrow x = 0, 2\pi, 4\pi, \frac{\pi}{2}, \frac{5\pi}{2}$$

$$\Rightarrow 5 \text{ solutions}$$

17. Ans (A)

$$0 \leq x^2 - x + 1 \leq 1$$

$$\Rightarrow x^2 - x \leq 0$$

$$\Rightarrow x \in [0, 1]$$

$$\text{Also, } 0 < \sin^{-1} \left(\frac{2x-1}{2} \right) \leq \frac{\pi}{2}$$

$$\Rightarrow 0 < \frac{2x-1}{2} \leq 1$$

$$\Rightarrow 0 < 2x-1 \leq 2$$

$$1 < 2x \leq 3$$

$$\frac{1}{2} < x \leq \frac{3}{2}$$

Taking intersection

$$x \in \left(\frac{1}{2}, 1\right]$$

$$\Rightarrow \alpha = \frac{1}{2}, \beta = 1$$

$$\Rightarrow \alpha + \beta = \frac{3}{2}$$

18. Ans (A)

For continuity

$$\lim_{x \rightarrow 0} \frac{x^3}{4\sin^4 x} (\ln(1 + 2xe^{-2x}) - 2\ln(1 - xe^{-x})) = \alpha$$

$$\lim_{x \rightarrow 0} \frac{1}{4x} [2xe^{-2x} + 2xe^{-x}] = \alpha$$

$$= \frac{1}{4}(4) = \alpha = 1$$

19. Ans (D)

Tangent to hyperbola of

Slope $m = -2$ (given)

$$y = -2x \pm \sqrt{3(3)}$$

$$(y = mx \pm \sqrt{a^2 m^2 - b^2})$$

$$\Rightarrow y + 2x = \pm 3 \Rightarrow 2x + y = 3 \quad (k > 0)$$

For parabola $y^2 = ax$

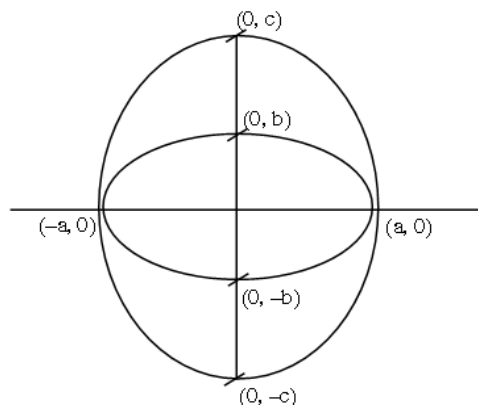
$$y = mx + \frac{a}{4m}$$

$$\Rightarrow y = -2x + \frac{a}{-8}$$

$$\Rightarrow \frac{a}{-8} = 3$$

$$\Rightarrow a = -24$$

20. Ans (A)



$$e^2 = 1 - \frac{b^2}{a^2}$$

$$e^2 = 1 - \frac{a^2}{c^2}$$

$$\Rightarrow \frac{b^2}{a^2} = \frac{a^2}{c^2}$$

$$\Rightarrow c^2 = \frac{a^4}{b^2} \Rightarrow c = \frac{a^2}{b}$$

Also $b = ce$

$$\Rightarrow c = \frac{b}{e}$$

$$\frac{b}{e} = \frac{a^2}{b}$$

$$\Rightarrow e = \frac{b^2}{a^2} = 1 - e^2$$

$$\Rightarrow e^2 + e - 1 = 0$$

$$\Rightarrow e = \frac{-1 + \sqrt{5}}{2}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (720)

$$f(1) + f(2) = 3 - f(3)$$

$$\Rightarrow f(1) + f(2) = 3 + f(3) = 3$$

The only possibility is : $0 + 1 + 2 = 3$

$\Rightarrow$ Elements 1, 2, 3 in the domain can be mapped with 0, 1, 2 only.

So number of bijective functions.

$$= |3| \times |5| = 720$$

2. Ans (96)

2	4	6	8				
---	---	---	---	--	--	--	--

4 4 3 2 1

$$= 4 \times 4 \times 3 \times 2 = 96$$

3. Ans (3125)

$$\text{Let matrix } B = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\therefore AB = BA$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} d & e & f \\ a & b & c \\ g & h & i \end{bmatrix} = \begin{bmatrix} b & a & c \\ e & d & f \\ h & g & i \end{bmatrix}$$

$$\Rightarrow d = b, e = a, f = c, g = h$$

$$\therefore \text{Matrix } B = \begin{bmatrix} a & b & c \\ b & a & c \\ g & g & i \end{bmatrix}$$

No. of ways of selecting a, b, c, g, i

$$= 5 \times 5 \times 5 \times 5 \times 5$$

$$= 5^5 = 3125$$

$$\therefore \text{No. of Matrices } B = 3125$$

4. Ans (4)

Class	Frequency	x_i	$f_i x_i$
0-6	a	3	3a
6-12	b	9	9b
12-18	12	15	180
18-24	9	21	189
24-30	5	27	135
	$N=(26+a+b)$		$(504+3a+9b)$

$$\text{Mean} = \frac{3a + 9b + 180 + 189 + 135}{a + b + 26} = \frac{309}{22}$$

$$\Rightarrow 66a + 198b + 11088 = 309a + 309b + 8034$$

$$\Rightarrow 243a + 111b = 3054$$

$$\Rightarrow \boxed{81a + 37b = 1018} \rightarrow (1)$$

$$\text{Now, Median} = 12 + \frac{\frac{a+b+26}{2} - (a+b)}{12} \times 6 = 14$$

$$\Rightarrow \frac{13}{2} - \left(\frac{a+b}{4} \right) = 2$$

$$\Rightarrow \frac{a+b}{4} = \frac{9}{2}$$

$$\Rightarrow \boxed{a+b = 18} \rightarrow (2)$$

From equation (1) & (2)

$$a = 8, b = 10$$

$$\therefore (a-b)^2 = (8-10)^2$$

5. Ans (1251)

$$2040 = 2^3 \times 3 \times 5 \times 17$$

n should not be multiple of 2, 3, 5 and 17.

$$\text{Sum of all } n = (1 + 3 + 5 + \dots + 99) - (3 + 9 +$$

$$15 + 21 + \dots + 99) - (5 + 25 + 35 + 55 + 65 +$$

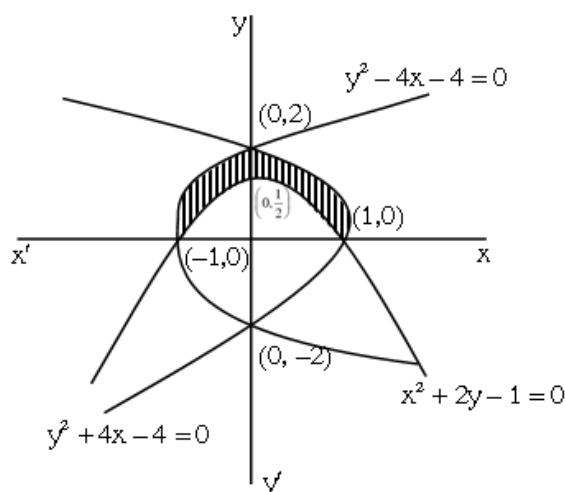
$$85 + 95) - (17)$$

$$= 2500 - \frac{17}{2}(3 + 99) - 365 - 17$$

$$= 2500 - 867 - 365 - 17$$

$$= 1251$$

6. Ans (2)



Required Area (shaded)

$$= 2 \left[\int_0^2 \left(\frac{4-y^2}{4} \right) dy - \int_0^1 \left(\frac{1-x^2}{2} \right) dx \right]$$

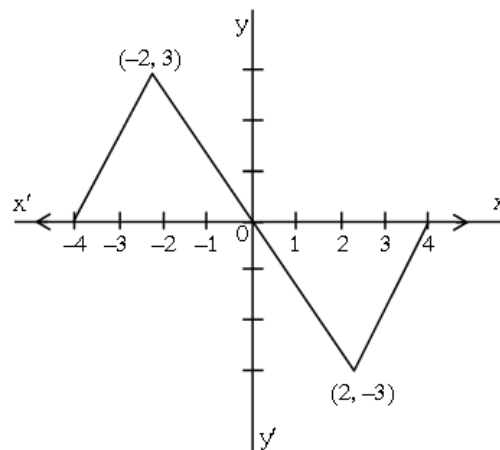
$$= 2 \left[\frac{4}{3} - \frac{1}{3} \right] = (2)$$

7. Ans (4)

$$f(x-2) = \begin{cases} \frac{3x}{2} & -4 \leq x \leq -2 \\ -\frac{3x}{2} & -2 < x \leq 0 \\ 0 & x \in (-\infty, -4) \cup (0, +\infty) \end{cases}$$

$$f(x-2) = \begin{cases} \frac{3x}{2} & 0 \leq x \leq 2 \\ -\frac{3x}{2} + 6 & 2 \leq x \leq 4 \\ 0 & x \in (-\infty, 0) \cup (4, +\infty) \end{cases}$$

$$g(x) = f(x+2) - f(x-2) = \begin{cases} \frac{3x}{2} + 6 & -4 \leq x \leq -2 \\ -\frac{3x}{2} & -2 < x < 2 \\ \frac{3x}{2} - 6 & 2 \leq x \leq 4 \\ 0 & x \in (-\infty, -4) \cup (4, \infty) \end{cases}$$



$$n = 0$$

$$m = 4 \Rightarrow (n + m = 4)$$

8. Ans (8)

$$\left(2x^r + \frac{1}{x^2} \right)^{10}$$

$$\text{General term} = {}^{10}C_R (2x^2)^{10-R} x^{-2R}$$

$$\Rightarrow 2^{10-R} {}^{10}C_R = 180 \dots\dots\dots (1)$$

$$\& (10 - R)r - 2R = 0$$

$$r = \frac{2R}{10 - R}$$

$$r = \frac{2(R - 10)}{10 - R} + \frac{20}{10 - R}$$

$$\Rightarrow r = -2 + \frac{20}{10 - R} \dots(2)$$

R = 8 or 5 reject equation (1) not satisfied

At R = 8

$$2^{10-R} {}^{10}C_R = 180 \Rightarrow \boxed{r = 8}$$

9. Ans (4)

$$y + 1 = Y \Rightarrow dy = dY$$

$$x + 2 = X \Rightarrow dx = dX$$

$$\Rightarrow \left(X e^{\frac{Y}{X}} + Y \right) dX = X dY$$

$$\Rightarrow X dY - Y dX = X e^{\frac{Y}{X}} dX$$

$$\Rightarrow d \left(\frac{Y}{X} \right) e^{-\frac{Y}{X}} = \frac{dX}{X}$$

$$-e^{-\frac{Y}{X}} = \ell |X| + c$$

$$(3, 2) \rightarrow -e^{-\frac{2}{3}} = \ell |3| + c$$

$$-e^{-\frac{Y}{X}} = \ell \ln |X| - e^{-\frac{2}{3}} - \ell \ln 3$$

$$e^{-\frac{Y}{X}} = e^{2/3} + \ell \ln 3 - \ell \ln |X| > 0$$

$$\ell \ln |X| < \left(e^{2/3} + \ell \ln 3 \right)$$

$$\text{Let } \lambda = \left(e^{2/3} + \ell \ln 3 \right)$$

$$|x + 2| < e^{\lambda}$$

$$-e^{\lambda} < x + 2 < e^{\lambda}$$

$$-e^{\lambda} - 2 < x < e^{\lambda} - 2$$

$$\alpha \quad \beta$$

$$\alpha + \beta = -4 \Rightarrow |\alpha + \beta| = 4$$

Although $x = -2$ should be excluded from domain but according to the given problem it will be the most appropriate solution.

10. Ans (96)

$$11^n > 10^n + 9^n$$

$$\Rightarrow 11^n - 9^n > 10^n$$

$$\Rightarrow (10 + 1)^n - (10 - 1)^n > 10^n$$

$$\Rightarrow \left\{ {}^nC_1 \cdot 10^{n-1} + {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots \right\} > 10^n$$

$$\Rightarrow 2n \cdot 10^{n-1} + 2 \left\{ {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots \right\} > 10^n \quad \dots (1)$$

For $n = 5$

$$10^5 + 2 \left\{ {}^5C_3 10^2 + {}^5C_5 \right\} > 10^5 \text{ (True)}$$

For $n = 6, 7, 8, \dots, 100$

$$2n 10^{n-1} > 10^n$$

$$\Rightarrow 2n 10^{n-1} + 2 \left\{ {}^nC_3 10^{n-3} + {}^nC_5 10^{n-5} + \dots \right\} > 10^n$$

$$\Rightarrow 11^n - 9^n > 10^n \text{ For } n = 5, 6, 7, \dots, 100$$

For $n = 4$, Inequality (1) is not satisfied

$\Rightarrow$ Inequality does not hold good for

$$N = 1, 2, 3, 4$$

So, required number of elements

$$= 96$$